

ZHENZHUO LI

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EDUCATION

Hong Kong University of Science and Technology (Guangzhou) BSc, Data Science and Big Data Technology 2023 – 2027

- **CGA:** 3.807/4.3 | **Major CGA:** 3.837 | Dean's List (3 semesters: 2023-24 Spring, 2024-25 Fall, 2024-25 Spring)
- **Core Coursework:** Machine Learning (A-), Deep Learning (B+), Introduction to AI (A+), Design and Analysis of Algorithms (A), Advanced Algorithms and Data Structures (A+), Database Management Systems (A+), Cloud Computing and Big Data Systems (A-), Applied Statistics (A-), Linear Algebra (A+), Discrete Mathematics (A), Honors Calculus I/II (A+/A+)
- **Exchange:** Osaka University, Japan (2025 Fall)

Zhengzhou Foreign Language School High School 2020 – 2023

- National Physics Olympiad (CPHO) — **Provincial First Prize**, 38th and 39th editions (consecutive)

RESEARCH EXPERIENCE

LENS: Large Pre-trained Transformer for Financial Time Series Co-author 2024.06 – 2024.10
Research Assistant | Supervisor: Prof. Guang Zhang, HKUST(GZ) FinTech

- Designed and implemented high-concurrency data preprocessing pipelines for TB-level financial trading data; achieved **13× speedup** on data cleaning and **6× speedup** on statistical aggregation through parallelization and memory-efficient chunking
- Paper accepted at **ACM ICAIF 2025** (co-author); also available as arXiv:2408.10111

SpatialReflect: Spatially Disentangled Reflection for Controllable Generation First Author
2025.06 – Present

Undergraduate Researcher | Supervisor: Prof. Zeke Xie, HKUST(GZ) DSA

- Identified three independent failure modes in MultiDiffusion-based multi-region generation; proposed SpatialReflect, a training-free framework with formal guarantees (3 theorems and 2 propositions)
- Submitted to **PRCV 2026** (First Author)

Multi-Agent Debate as Verifiable Reasoning Ongoing Exploratory Research 2026.06 – Present
Self-proposed; in discussion with potential advisors

- Investigating the reliability of multi-agent LLM debate as a verification mechanism: formalizing error decay bounds under agent heterogeneity, convergence properties, and calibrated confidence estimation
- Preliminary theoretical framework documented; exploring applications to financial research agents where multi-agent reasoning can improve signal reliability and reduce individual model bias

PUBLICATIONS

- Y. Xu, J. Hao, A. Liu, **Z. Li**, S. Meng, S. Yuan, G. Zhang, “LENS: Large Pre-trained Transformer for Exploring Financial Time Series Regularities,” in *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF 2025)*, pp. 771–778, 2025.
- Y. Xu, A. Liu, J. Hao, **Z. Li**, S. Meng, G. Zhang, “Plutus: A Well Pre-trained Large Unified Transformer Can Unveil Financial Time Series Regularities,” *arXiv preprint arXiv:2408.10111*, 2024.
- **Z. Li**, et al., “SpatialReflect: Spatially Disentangled Reflection for Multi-Region Style-Controlled Generation,” submitted to *PRCV 2026*. (First Author)

AWARDS & HONORS

- **ASC World University Supercomputer Competition 2026 — First Prize.** Contributed to the UniFOLM-WMA track: I/O optimization for large-model inference and Qibotn quantum circuit simulation backend and algorithm optimization.
- **INFH Lizhi Innovation Talent Scholarship, HKUST(GZ)** — Awarded for innovative research potential (2025)
- **Dean’s List, HKUST(GZ)** — Awarded for academic excellence in the top percentile (3 semesters)
- **MCM/ICM Mathematical Contest in Modeling** — Honorable Mention (2025)
- **ASC World University Supercomputer Competition 2025** — Second Prize (Team Member)
- **Admission Scholarship (First Class), HKUST(GZ)** — Merit-based scholarship (2023)

PROJECT EXPERIENCE

AI-Assisted Financial Research and Signal Validation Infrastructure Independent 2026.01 – 2026.06

Independent Developer | Hermes Agent + LangGraph Framework

- Built a research-oriented financial signal pipeline: OSINT multi-source data collection (social media, financial forums) → LLM-based claim extraction → multi-factor screening → transaction-cost-aware backtesting with walk-forward validation
- Designed multi-agent orchestration with LangGraph StateGraph: Evidence Engine implementing Bayesian belief update, Physical Model Kernel for domain-constrained inference, and layered observation schema (L0–L6 source reliability tiering)
- Developed a unified factor library spanning 7 categories (price-volume, momentum, volatility, technical, moving-average, microstructure, regime) with automated IC analysis and factor decay monitoring
- Implemented ML ensemble framework: independent LightGBM signal generators (PriceVol, Momentum, Reversal, Sector RS) with probability-weighted combination; final XGBoost model with 80 engineered features
- Validated on 3-year historical window (2023–2025) with walk-forward optimization and regime-aware position sizing; tracked transaction costs and slippage throughout

SELECTED PROJECTS

Factor Research and Backtesting Module Sub-module of above 2026.01 – Present

- Built a full-stack factor research pipeline: factor computation (39 factors) → IC analysis (Spearman rank correlation) → factor decay monitoring → automated screening with 99% discard rate
- Implemented Fisher 15-point qualitative evaluation; Backtrader-based backtesting with walk-forward optimization and regime-aware position sizing
- Evaluated a three-tier position-sizing scheme (safety/stabilizer/accelerator) with signal decay monitoring

UniFOLM-WMA Inference and Simulation Optimization ASC2026 Competition Task 2026.01 – 2026.05

- Contributed I/O optimization for large-model inference on NVIDIA A800 cluster; co-developed Qibotn quantum circuit simulation backend and algorithm optimization

TECHNICAL SKILLS

Programming & Data: Python, C++, SQL; NumPy, Pandas, SciPy, scikit-learn
AI/ML: PyTorch, Transformers, LightGBM, XGBoost, LangGraph, LangChain
Financial ML: Factor modeling, IC/decay analysis, walk-forward backtesting, transaction-cost-aware evaluation, regime-aware risk control
Systems: Linux, Git, Docker, CUDA, MPI, Slurm, MySQL/PostgreSQL
Languages: Chinese (native), English (academic working proficiency), Japanese (JLPT N1)

Research Interests: AI for Financial Markets and Time-Series Forecasting; Verifiable Multi-Agent Systems for Financial Research; LLM Reliability and Calibration under Domain Feedback